METRIC

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01 March 2007

DEPARTMENT OF DEFENSE
STANDARD PRACTICE

# CHROMIUM PLATING, LOW EMBRITTLEMENT, ELECTRODEPOSITION

AMSC N/A

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# F O R E W O R D

1. This military standard is approved for use by 309 MXSG/MXRL, Department of the Air Force and is available for use by all departments and agencies of the Department of Defense.

2. This standard provides guidance for the Air Force repair process, acquisition, and manufacture of parts and/or spare parts on the landing gear of all military aircraft.

3. Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: 309 MXSG/MXRL, Hill AFB, UT 84056-2609 or e-mailed to: 309MXSUG/MXRL@hill.af.mil. Since contact information can change, verification of currency of this address information through ASSIST Online database at https://assist.daps.dla.mil.



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# 1. SCOPE

1.1 Scope This standard covers the engineering requirements for electrodeposition of hard chromium on high strength steel substrates and the properties of deposit. Subsequent heat treating techniques needed to insure low hydrogen embrittlement of steel is described.

1.2 Documentation Chromium plated to this standard will meet the performance requirements of SAE-AMS-2460, which supersedes SAE-AMS-QQ-C-320. This standard may be used whenever SAE-AMS-QQ-C-320 is specified.

1.3 Classification

1.3.1 Classes. The following classes of chromium plating covered by this standard:

a. Class 1 – Crack free (see para.4.9)

b. Class 2 – Limited cracking allowed (see para.4.10)

c. Class 3 – Moderate cracking allowed (see para.4.11)

NOTE: When not specified, Class 3 plating will be used.

1.3.2 Types. The following types of chrome plating covered by this standard:

a. Type I – Plated to specified dimensions (see para.4.12)

b. Type II – Processed to specified dimensions after plating (see para.4.13)

# 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1. Specifications, standards, handbooks, and commercial item descriptions. The following specifications, standards, handbooks, and commercial item descriptions form a part of this document to the extent specified herein. Unless otherwise specified the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

| | |
|---|---|
| A-A-55827 | Chromium Trioxide, Technical |
| A-A-55828 | Sulfuric Acid, Technical |



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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-R-81841 Rotary Flap Peening of Metal Parts

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-866 Grinding of Chrome Plated Steel and Steel Parts Heat Treated to 180,000 PSI or Over

MIL-STD-871 Electro-Chemical Stripping of Inorganic Finishes (Inactive)

MIL-STD-1504 Abrasive Blast of Aircraft Components (Inactive)

(Copies of these documents are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094 or https://assist.daps.dla.mil.)

2.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)
29 CFR 1910 Toxic and Hazardous Substances

(Copies of this document are available online at http://www.gpoaccess.gov/cfr/index.html or from the Superintendent of Documents, U.S. Printing Office, North Capitol & "H" Streets, N.W., Washington D.C. 20402-0002.)

2.3 Non-Government publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated, the current revision in effect on date of invitation for bids or request for proposal is applicable.

Society of Automotive Engineers (SAE) International

SAE-AMS2430 Shot Peening

SAE-AMS2644 Inspection Material, Penetrant

SAE-AMS-QQ-C320 Chromium Plating (Electrodeposited)

(Copies of this document are available online at www.sae.org or from SAE World Headquarters, 400 Commonwealth Drive, Warrendale, PA 15096-0001)



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American Society for Testing and Materials (ASTM)

ASTM E1444 Magnetic Particle Testing

ASTM E1417 Standard Practice for Liquid Penetrate Examination

(Application for copies of these documents should be addressed to ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959 or http://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. DEFINITIONS

3.1 High strength steel. For the purpose of this standard high strength steel is defined as steel heat treated to 1.24 X 10° Pa (180,000 psi) and above.

3.2 Material batch. All items processed at one time through the plating bath.

## 4. GENERAL REQUIREMENTS

4.1 General. Chrome plating is applied for wear resistance, reconditioning worn or undersize parts, and such incidental corrosion protection is desired, an undercoat of nickel 0.025 to 0.050 millimeters (0.001 to 0.002 inches) thickness is recommended.

4.2 Materials and equipment.

4.2.1 Materials.

a. Chromium Trioxide, Technical (A-A-55827)

b. Sulfuric Acid, Technical (A-A-55828)

c. Anodes

7% Tin-lead, wires, rod, or strip

7% Antimony-lead, wires, rod, or strip

4.2.2 Equipment.



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a. Power source. Either generated or rectified D.C. current may be used. Ripple value shall not exceed 10 % (percent) as measured by dividing the Root Mean Square of the A.C. voltage component by the D.C. voltage. This can best be measured by using an RMS A.C. voltage meter and dividing this value by the D.C. voltage. These measurements are to be taken across the anode and cathode bus at the tank.

b. Tanks. Tanks shall be resistant to the operating temperature and the chemical environment.

c. Temperature control. Plating tanks to be operated at temperatures other than room temperature shall be equipped with automatic temperature indicating and regulating devices.

d. Instrumentation. An ammeter shall be placed in series with the chromium tank cathode. The ammeter shall have sufficient shuts and switches to provide a full-scale reading equal to the maximum capacity of the power source, and an accuracy of ± 5 percent of the current being measured.

e. Blast equipment. Blast cabinets shall be located near the plating line. The size of the cabinet shall be adequate to enclose the parts to be plated. Air lines shall be suitably trapped and filtered to prevent in-process contamination of the parts to be cleaned.

f. Bake oven. An oven capable of baking parts at 191 ± 14°C (375 ± 25°F) shall be located near the plating line. The size of the oven shall be adequate to enclose parts to be plated. The oven shall be equipped with temperature indicating, recording and regulating devices.

4.3 Specification SAE-AMS-QQ-C-320. All parts shall be plated in accordance with the requirements of SAE-AMS-QQ-C-320 and in conjunction with those specified in this standard. If there is a conflict between the two documents; however, the requirements of this document shall govern.

4.4 Finish. The plated part will have a finish that is continuous, homogenous, adherent, and free from harmful defects.

4.5 Shot peening. All parts shall be shot peened in accordance with SAE-AMS2430 and MIL-R-81841, unless otherwise specified.

4.6 Embrittlement. Qualification test specimens and process control test specimens shall be subjected to a sustained load test at 75 percent of the ultimate notched tensile strength. The specimens shall endure this sustained load for 200 hours minimum without failing or cracking.

4.7 Reprocessing. Parts rejected for defective plating, requiring stripping and re-plating, shall include all of the pre-plating steps of this standard. Parts shall be



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stripped in accordance with MIL-STD-871.

4.8 Plating thickness. The plating thickness shall be as specified on the engineering drawing or other applicable directives. Except when otherwise specified, the minimum thickness shall be 0.05 millimeters (0.002 inches) and the maximum thickness 0.2 millimeters (0.008 inches) on the finished part.

4.9 Class 1. Parts plated to Class 1 shall show no mud cracking pattern when fluorescent penetrant inspected (see para.5.3.1).

4.10 Class 2. Parts plated to Class 2 are allowed to show some cracking pattern when fluorescent penetrant inspected (see para.5.3.1). As a general rule cracks exceeding 13 millimeters (1/2 inch) in length or individual areas of cracking exceeding 19 square centimeters (3 square inches) or 10 percent of plated area, whichever is smaller, will be cause for rejection of the plated area.

4.11 Class 3. Parts plated to Class 3 are allowed to show a moderate amount of cracking when fluorescent penetrant inspected (see para.5.3.1). To determine part acceptability the part will be cleaned to remove the penetrant materials and the areas of cracking inspected visually using an oblique light. If the cracking pattern can be discerned, that will be cause for rejection of the plated area.

4.12 Type I plating. For Type I plating, the item shall be plated to the dimension and surface finish specified on the drawing. The surface finish on the item before plating shall be equal to or better than the required finish after plating. Type I plating may be buffed or lapped after plating if dimensional tolerances and surface conditions cannot be controlled in the plating operation.

4.13 Type II plating. For Type II plating a minimum of 0.05 millimeters (0.002 inch) more chrome than desired shall be deposited (per surface). The excess shall be ground off to give the final dimension and surface finish desired. Steel parts heat treated to 1.24 X 109 Pa (180,000 psi) and above shall be ground in accordance with MIL-STD-866. All ground chrome plated surfaces shall be fluorescent penetrant inspected (see 5.3.1). Chrome plated surfaces showing indications of abusive grinding (spiral, barber pole, circular, patch or linear crack patterns), spalling or blistering shall be cause for rejection of the plated area.

## 5. DETAILED REQUIREMENTS

5.1 Individual tasks, requirements, or test methods.

5.1.1 Prior to plating. Except for finish grinding operations, all machining, forming, welding and shot peening shall be completed prior to plating.

5.1.2 Baking. Parts shall be baked for stress relief before plating for four hours minimum at 191 ± 14°C (375 ± 25°F). Shot peening shall be performed before plating



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and after stress relieving.

5.1.3 Plating sequence. If chromium and cadmium are used in combination, the chromium shall be deposited first. When chrome plating is to be followed by cadmium plating, the 23 hours minimum bake period, required by para.5.2.7, can be replaced by a four hours bake period at 191 ± 14°C (375 ± 25°F), provided the part is baked for 23 hours minimum after completion of the cadmium plating.

5.1.4 Storage of parts. Storage of parts between stress relief and cleaning shall be controlled to prevent contact with water or other corrosive materials. Parts shall be stored to permit free circulation of air around the parts.

5.1.5 Handling of parts. After the parts have been cleaned, they shall be handled in such a manner that will assure a minimum of contamination.

5.1.6 Masking. Sections or areas of a part that are not to be plated shall be masked off. Plug and masking materials that do not contaminate the plating bath shall be used. Masking shall be performed at the most convenient step prior to plating.

5.1.7 Racking. Sufficient contact area and pressure shall be provided to carry the current without overheating. Racking should be performed at the most convenient step prior to plating. When gang plating (two or more like parts) care should be taken to assure a uniform division of current flow to each part by uniform tank spacing and meticulous cleaning and shaping of racking contact surfaces or by providing isolated controlled current paths to each individual part; while maintaining the amperage limits specified (see paragraph 5.2.4).

5.2 Plating Procedure. The chromium plating procedure shall be as described below:

5.2.1 Step No. 1. Parts shall be vapor degreased, solvent cleaned or alkaline cleaned. Cathodic cleaning is not permitted.

**WARNING**
**PROVIDE ADEQUATE VENTILATION DURING DEGREASING OPERATIONS. AVOID SKIN AND EYE CONTACT WITH SOLVENT SOLUTIONS.**

5.2.2 Step No. 2. The preferred method of cleaning is by dry blasting using 80-320 grit aluminum oxide (Al2O3), silicon dioxide (SiO2), garnet per MIL-STD-1504, or hand sanding. Other non-embrittling cleaning processes can be used with the approval of the procuring activity. Elapsed time between completion of cleaning and Step No. 3 shall not exceed Four (4) hours.

5.2.3 Step No. 3. Anodic etch 15.5 to 46.5 amps/decimeter squared (1 to 3A per square inch) for 30 seconds to 10 minutes. Etch in the following solution (preferred)



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or in the plating bath, per this paragraph:

Chromic Acid: 225-300 gram/liter (30-40 ounce/gallon)

Temperature: 32-60°C (90-140°F).

**CAUTION:** Etching increases iron contamination in the plating bath; therefore, the use of the above solution is recommended.

5.2.3.1 Alternate reverse etch solutions. A sulfuric acid or a sulfuric-hydrofluoric acid mixture can be used as an alternate, reverse etch solution, to the solution specified in paragraph 5.2.3. The conditions for the employment of either the sulfuric acid or sulfuric-hydrofluoric acid mixtures are as follows:

a. Forty (40) percent sulfuric acid:

(1) Operating range – Concentrated (93%) sulfuric acid diluted to 35 to 45% by volume.

(2) Temperature – ambient.

(3) Current density – 31.0 to 62.0 amps/decimeter squared (2 to 4A per square inch), anodic.

(4) Time – 30-120 seconds.

b. Sulfuric-hydrofluoric acid.

(1) Operating range – Concentrated (93%) sulfuric acid diluted to 20 to 30% by volume and concentrated (70%) hydrofluoric acid diluted to 2 to 7% by volume. Ammonium bifluoride may be used at an equivalent total fluoride concentration in place of hydrofluoric acid.

(2) Temperature – ambient.

(3) Current density – 31.0 to 62.0 amps/decimeter squared (2 to 4A per square inch), anodic.

(4) Time – 60-90 seconds.

c. Twenty-five percent (25%) sulfuric acid.

(1) Operating range – Concentrated (93%) sulfuric acid diluted to 20 to 30% sulfuric acid by volume.

(2) Temperature – Ambient



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(3) Current density 31.0 to 62.0 amps/decimeter squared (2 to 4A per square inch) anodic.

(4) Time – 30 – 120 seconds.

5.2.4 Step No. 4. Chromium plate at 15.5 to 77.5 amps/decimeters squared (1 to 5A per square inch) to the required thickness. When feasible, apply voltage so that the current will flow upon immersion of parts. Chromium plate in the following solutions:

a. Chromic Acid 225-270 gram/liter (30-36 ounce/gallon)

b. Sulfuric Acid 2.25-2.7 gram/liter (0.30-0.36 ounce/gallon)
c. Trivalent Chrome 4 grams per liter (g/l) maximum

d. Iron 5 g/l maximum

e. Ratio Chromic Acid/Sulfuric Acid 80/1 to 120/1

f. Temperature 52 - 62°C (125 - 145°F)

g. It is general practice to start plating at a higher current (two to three times the plating current) for a short time (approximately 0.5 to 1.0 minutes) and then reducing to the desired plating current. This helps in providing the initial layer of chromium and lessens the chance of leaving bare spots.

NOTE: Other plating solutions may be used when approved by the responsible engineering organization provided the deposit meets the requirements of this standard.

**WARNING**
**PROVIDE ADEQUATE VENTILATION WHEN USING CHROMIC PLATING SOLUTIONS. WEAR PROTECTIVE RUBBER GLOVES AND CHEMICAL GOOGLES TO PREVENT SKIN AND EYE EXPOSURE.**

5.2.5 Step No. 5. Rinse parts in cold water and inspect for defects and adequate buildup. If undersize, return to plating tank and continue plating. If plating is adequate, remove masking, racking, and clean as necessary.

5.2.6 Step No. 6. Rinse all parts in hot water and blow dry with compressed air. Elapsed time between completion of plating and start of baking, Step No. 7, shall not exceed four hours.

5.2.7 Step No. 7. Bake all parts heat treated to 1.24 X 109 Pa (180,000 psi) and above for twenty three hours minimum, at 191 ± 14°C (375 ± 25°F) (See paragraph 5.1.3).

5.3 Inspection. Inspection shall be in accordance with the inspection and tests



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in AMS-QQ-C-320 and this standard.

5.3.1 Penetrant inspection. The parts shall be cleaned and inspected for the requirements of paragraphs 4.9, 4.10, 4.11 and 4.13 in accordance with ASTM-E1417, Type I, Method A, B, and C using materials per SAE-AMS-2644 Group VI. Penetrant penetration time will be 20 minutes minimum.

5.3.2 Magnetic particle inspection. When specified, parts shall be cleaned and inspected per ASTM-E1444, for any deleterious processing effects on the part. The acceptance standard should be as required by responsible engineering organization.

5.4 Process Controls. Solutions used in the plating process shall be checked periodically and maintained in accordance with the requirements of this process standard

5.5 Qualification Embrittlement Test. The vendor shall demonstrate his ability to provide chromium plate, which meets the requirements of para.4.6 as follows:

a. Four round notched 4340 steel specimens per ASTM F 519, Type 1a.1 or 1a.2 shall be prepared.

b. The specimens shall be prepared for plating and plated in accordance with all of the requirements of this specification. During plating the specimens shall be mounted symmetrically on a rack by themselves. All areas of the rack except the contact area shall be coated with a suitable maskant. An ammeter, having a sensitivity of 0.5 amperes or better shall be connected in the plating circuit near the specimens being plated. The ammeter shall either be built into the rectifier being used or shall be inserted between the rectifier and the cathode. The specimen shall be plated at two amperes per square inch for three hours. The specimen shall be baked for twenty-three hours at 191 ± 14°C (375 ± 25°F) within four hours of removal from the bath.

c. The specimens will be subjected to 200 hours of static loading at 75 percent of the ultimate notched tensile strength. The test shall be considered passed if all four specimens meet the requirements of paragraph 4.6.

d. Upon successful completion of the static load test, one of the notched tensile specimens shall be sectioned across the notch parallel to the axis of the specimen. Photomicrographs shall be taken of the notched area and examined for complete coverage of the notch (use 80 – 100X magnification).

5.6 Process control embrittlement acceptance test. The process control embrittlement acceptance test shall be as follows:

a. Two standard specimens of the type noted in paragraph 5.5.a shall be plated per paragraph 5.5.b in conjunction with the plating of items. The specimens shall be subjected to a sustained load test of 75 percent of the ultimate notch tensile strength



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of the material for 200 hours minimum and shall meet the requirements of paragraph 4.6. Failure of any one of the specimens shall constitute failure of the test, and production shall cease until the cause of the failure is determined and the bath requalified. Acceptance of items completed after the last successfully completed acceptance test shall be withheld until the extent and cause of the failure have been determined.

b. The test for embrittlement shall be conducted as often as deemed necessary with a maximum interval of every ninety days. Prior to processing with a new bath, the bath must be qualified in accordance with paragraph 5.5.

5.7 Safety and health. This document specifies the use of certain materials, which have been listed in 29 CFR 1910 (OSHA Standards) as "Toxic and Hazardous Substances." Personnel exposure of these materials during the process must be limited to the values specified in applicable portions of the OSHA Standard 1910.1000.

## 6. NOTES

(This section contains information of a general or explanatory in nature that may be helpful, but is not mandatory.)

6.1 Intended use. This standard provides guidance on the procedures and processes for low embrittlement chromium electro-deposition for verification of the Air Force repair process on the landing gear of all aircrafts. It is an Air Force unique manufacturing standard that will be used only the Air Force at Hill Air Force Base.

6.2 Plating options. Chrome plating is applied for wear resistance, reconditioning worn or undersized parts, and such incidental corrosion protection as the specified thickness of the deposit may afford. Where corrosion protection is desired, and undercoat of nickel 0.025 to 0.050 millimeters (0.001 to 0.002 inches) thickness is recommended. When not specified, Class 3 plating is preferred (see paragraph 1.2.1).

6.3 Subject term (key word) listing.

Chrome
Corrosion
Process control

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.



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## CONCLUDING MATERIAL

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Agent Activity:
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